

## LPDDR2-S4 Refresh Requirement Parameters

| Parameter                                                                                        |       | Symbol  | 512Mb         | Unit |
|--------------------------------------------------------------------------------------------------|-------|---------|---------------|------|
| Number of Banks                                                                                  |       |         | 4             |      |
| Refresh Window $T_{case} \leq 85^{\circ}\text{C}$                                                |       | tREFW   | 32            | ms   |
| Refresh Window $85^{\circ}\text{C} < T_{case} \leq 105^{\circ}\text{C}$                          |       | tREFW   | 8             | ms   |
| Required number of REFRESH commands (min)                                                        |       | R       | 4096          |      |
| Average time between REFRESH commands (for reference only)<br>$T_{case} \leq 85^{\circ}\text{C}$ | REFab | tREFI   | 7.8           | us   |
|                                                                                                  | REFpb | tREFIpb | Not permitted |      |
| Refresh Cycle time                                                                               |       | tRFCab  | 90            | ns   |
| Per Bank Refresh Cycle time                                                                      |       | tRFCpb  | N/A           | ns   |
| Burst Refresh Window<br>= $(4 \times 8 \times tRFCab)$                                           |       | tREFBW  | 2.88          | us   |

## AC TIMINGS

LPDDR2 AC Timing Table (1,2)

| Parameter                                                                          | Symbol                              | min max    | min t <sub>CK</sub> | LPDDR2 |      |      |      |      |                                                                                                                                               |      |      |      |      | Unit                  |
|------------------------------------------------------------------------------------|-------------------------------------|------------|---------------------|--------|------|------|------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------|-----------------------|
|                                                                                    |                                     |            |                     | 1066   | 933  | 800  | 667  | 533  | 466                                                                                                                                           | 400  | 333  | 266  | 200  |                       |
| Max. Frequency                                                                     |                                     | ~          |                     | 533    | 466  | 400  | 333  | 266  | 233                                                                                                                                           | 200  | 166  | 133  | 100  | MHz                   |
| Clock Timing                                                                       |                                     |            |                     |        |      |      |      |      |                                                                                                                                               |      |      |      |      |                       |
| Average Clock Period                                                               | t <sub>CK</sub> (avg)               | min<br>max |                     |        |      | 2.5  | 3    | 3.75 | 4.3                                                                                                                                           | 5    | 6    | 7.5  | 10   | ns                    |
| Average high pulse width                                                           | t <sub>CH</sub> (avg)               | min        |                     |        |      |      |      |      | 0.45                                                                                                                                          |      |      |      |      | t <sub>CK</sub> (avg) |
|                                                                                    |                                     | max        |                     |        |      |      |      |      | 0.55                                                                                                                                          |      |      |      |      |                       |
| Average low pulse width                                                            | t <sub>CL</sub> (avg)               | min        |                     |        |      |      |      |      | 0.45                                                                                                                                          |      |      |      |      | t <sub>CK</sub> (avg) |
|                                                                                    |                                     | max        |                     |        |      |      |      |      | 0.55                                                                                                                                          |      |      |      |      |                       |
| Absolute Clock Period                                                              | t <sub>CK</sub> (abs)               | min        |                     |        |      |      |      |      | t <sub>CK</sub> (avg)min + t <sub>JIT</sub> (per),min                                                                                         |      |      |      |      | ps                    |
| Absolute clock HIGH pulse width<br>(with allowed jitter)                           | t <sub>CH</sub> (abs),<br>allowed   | min        |                     |        |      |      |      |      | 0.43                                                                                                                                          |      |      |      |      | t <sub>CK</sub> (avg) |
|                                                                                    |                                     | max        |                     |        |      |      |      |      | 0.57                                                                                                                                          |      |      |      |      | t <sub>CK</sub> (avg) |
| Absolute clock LOW pulse width<br>(with allowed jitter)                            | t <sub>CL</sub> (abs),<br>allowed   | min        |                     |        |      |      |      |      | 0.43                                                                                                                                          |      |      |      |      | t <sub>CK</sub> (avg) |
|                                                                                    |                                     | max        |                     |        |      |      |      |      | 0.57                                                                                                                                          |      |      |      |      | t <sub>CK</sub> (avg) |
| Clock Period Jitter (with allowed jitter)                                          | t <sub>JIT</sub> (per),<br>allowed  | min        |                     | -90    | -95  | -100 | -110 | -120 | -130                                                                                                                                          | -140 | -150 | -180 | -250 | ps                    |
|                                                                                    |                                     | max        |                     | 90     | 95   | 100  | 110  | 120  | 130                                                                                                                                           | 140  | 150  | 180  | 250  |                       |
| Maximum Clock Jitter between two<br>consecutive clock cycles (with allowed jitter) | t <sub>JIT</sub> (cc),<br>allowed   | max        |                     | 180    | 190  | 200  | 220  | 240  | 260                                                                                                                                           | 280  | 300  | 360  | 500  | ps                    |
| Duty cycle Jitter (with allowed jitter)                                            | t <sub>JIT</sub> (duty),<br>allowed | min        |                     |        |      |      |      |      | min((t <sub>CH</sub> (abs),min - t <sub>CH</sub> (avg),min), (t <sub>CL</sub> (abs),min - t <sub>CL</sub> (avg),min)) * t <sub>CK</sub> (avg) |      |      |      |      | ps                    |
|                                                                                    |                                     | max        |                     |        |      |      |      |      | max((t <sub>CH</sub> (abs),max - t <sub>CH</sub> (avg),max), (t <sub>CL</sub> (abs),max - t <sub>CL</sub> (avg),max)) * t <sub>CK</sub> (avg) |      |      |      |      | ps                    |
| Cumulative error across 2 cycles                                                   | t <sub>ERR</sub> (2per),<br>allowed | min        |                     | -132   | -140 | -147 | -162 | -177 | -191                                                                                                                                          | -206 | -221 | -265 | -368 | ps                    |
|                                                                                    |                                     | max        |                     | 132    | 140  | 147  | 162  | 177  | 191                                                                                                                                           | 206  | 221  | 265  | 368  |                       |
| Cumulative error across 3 cycles                                                   | t <sub>ERR</sub> (3per),<br>allowed | min        |                     | -157   | -166 | -175 | -192 | -210 | -227                                                                                                                                          | -245 | -262 | -314 | -437 | ps                    |
|                                                                                    |                                     | max        |                     | 157    | 166  | 175  | 192  | 210  | 227                                                                                                                                           | 245  | 262  | 314  | 437  |                       |
| Cumulative error across 4 cycles                                                   | t <sub>ERR</sub> (4per),<br>allowed | min        |                     | -175   | -185 | -194 | -214 | -233 | -253                                                                                                                                          | -272 | -291 | -350 | -486 | ps                    |
|                                                                                    |                                     | max        |                     | 175    | 185  | 194  | 214  | 233  | 253                                                                                                                                           | 272  | 291  | 350  | 486  |                       |
| Cumulative error across 5 cycles                                                   | t <sub>ERR</sub> (5per),<br>allowed | min        |                     | -188   | -199 | -209 | -230 | -251 | -272                                                                                                                                          | -293 | -314 | -377 | -524 | ps                    |
|                                                                                    |                                     | max        |                     | 188    | 199  | 209  | 230  | 251  | 272                                                                                                                                           | 293  | 314  | 377  | 524  |                       |
| Cumulative error across 6 cycles                                                   | t <sub>ERR</sub> (6per),<br>allowed | min        |                     | -200   | -211 | -222 | -244 | -266 | -288                                                                                                                                          | -311 | -333 | -399 | -555 | ps                    |
|                                                                                    |                                     | max        |                     | 200    | 211  | 222  | 244  | 266  | 288                                                                                                                                           | 311  | 333  | 399  | 555  |                       |
| Cumulative error across 7 cycles                                                   | t <sub>ERR</sub> (7per),<br>allowed | min        |                     | -209   | -221 | -232 | -256 | -279 | -302                                                                                                                                          | -325 | -348 | -418 | -581 | ps                    |
|                                                                                    |                                     | max        |                     | 209    | 221  | 232  | 256  | 279  | 302                                                                                                                                           | 325  | 348  | 418  | 581  |                       |

LPDDR2 AC Timing Table (1,2)

| Parameter                                                 | Symbol                               | min max | min t <sub>CK</sub> | LPDDR2                                                                                     |      |      |      |      |      |      |      |      |      | Unit |
|-----------------------------------------------------------|--------------------------------------|---------|---------------------|--------------------------------------------------------------------------------------------|------|------|------|------|------|------|------|------|------|------|
|                                                           |                                      |         |                     | 1066                                                                                       | 933  | 800  | 667  | 533  | 466  | 400  | 333  | 266  | 200  |      |
| Cumulative error across 8 cycles                          | t <sub>ERR</sub> (8per),<br>allowed  | min     |                     | -217                                                                                       | -229 | -214 | -266 | -290 | -314 | -338 | -362 | -435 | -604 | ps   |
|                                                           |                                      | max     |                     | 217                                                                                        | 229  | 214  | 266  | 290  | 314  | 338  | 362  | 435  | 604  |      |
| Cumulative error across 9 cycles                          | t <sub>ERR</sub> (9per),<br>allowed  | min     |                     | -224                                                                                       | -237 | -249 | -274 | -299 | -324 | -349 | -374 | -449 | -624 | ps   |
|                                                           |                                      | max     |                     | 224                                                                                        | 237  | 249  | 274  | 299  | 324  | 349  | 374  | 449  | 624  |      |
| Cumulative error across 10 cycles                         | t <sub>ERR</sub> (10per),<br>allowed | min     |                     | -231                                                                                       | -244 | -257 | -282 | -308 | -334 | -359 | -385 | -462 | -641 | ps   |
|                                                           |                                      | max     |                     | 231                                                                                        | 244  | 257  | 282  | 308  | 334  | 359  | 385  | 462  | 641  |      |
| Cumulative error across 11 cycles                         | t <sub>ERR</sub> (11per),<br>allowed | min     |                     | -237                                                                                       | -250 | -263 | -289 | -316 | -342 | -368 | -395 | -474 | -658 | ps   |
|                                                           |                                      | max     |                     | 237                                                                                        | 250  | 263  | 289  | 316  | 342  | 368  | 395  | 474  | 658  |      |
| Cumulative error across 12 cycles                         | t <sub>ERR</sub> (12per),<br>allowed | min     |                     | -242                                                                                       | -256 | -269 | -296 | -323 | -350 | -377 | -403 | -484 | -672 | ps   |
|                                                           |                                      | max     |                     | 242                                                                                        | 256  | 269  | 296  | 323  | 350  | 377  | 403  | 484  | 672  |      |
| Cumulative error across n = 13, 14 . . .<br>49, 50 cycles | t <sub>ERR</sub> (nper),<br>allowed  | min     |                     | t <sub>ERR</sub> (nper),allowed,min = (1 + 0.68ln(n)) * t <sub>JIT</sub> (per),allowed,min |      |      |      |      |      |      |      |      |      | ps   |
|                                                           |                                      | max     |                     | t <sub>ERR</sub> (nper),allowed,max = (1 + 0.68ln(n)) * t <sub>JIT</sub> (per),allowed,max |      |      |      |      |      |      |      |      |      |      |

LPDDR2 AC Timing Table (1,2)

| Parameter                            | Symbol               | min max | min t <sub>CK</sub> | LPDDR2                                                     |      |      |      |      |      |      |      |      |      | Unit                  |
|--------------------------------------|----------------------|---------|---------------------|------------------------------------------------------------|------|------|------|------|------|------|------|------|------|-----------------------|
|                                      |                      |         |                     | 1066                                                       | 933  | 800  | 667  | 533  | 466  | 400  | 333  | 266  | 200  |                       |
| ZQ Calibration Parameters            |                      |         |                     |                                                            |      |      |      |      |      |      |      |      |      |                       |
| Initialization Calibration Time      | t <sub>ZQINIT</sub>  | min     |                     | 1                                                          |      |      |      |      |      |      |      |      |      | us                    |
| Long Calibration Time                | t <sub>ZQCL</sub>    | min     | 6                   | 360                                                        |      |      |      |      |      |      |      |      |      | ns                    |
| Short Calibration Time               | t <sub>ZQCS</sub>    | min     | 6                   | 90                                                         |      |      |      |      |      |      |      |      |      | ns                    |
| Calibration Reset Time               | t <sub>ZQRESET</sub> | min     | 3                   | 50                                                         |      |      |      |      |      |      |      |      |      | ns                    |
| Read Parameters (3)                  |                      |         |                     |                                                            |      |      |      |      |      |      |      |      |      |                       |
| DQS output access time from CK/CK#   | t <sub>DQSCK</sub>   | min     |                     | 2500                                                       |      |      |      |      |      |      |      |      |      | ps                    |
|                                      |                      | max     |                     | 5500                                                       |      |      |      |      |      |      |      |      |      |                       |
| DQSCK Delta Short <sup>(4)</sup>     | t <sub>DQSCKDS</sub> | max     |                     | 330                                                        | 380  | 450  | 540  | 670  | 770  | 900  | 1080 | 1350 | 1800 | ps                    |
| DQSCK Delta Medium <sup>(5)</sup>    | t <sub>DQSCKDM</sub> | max     |                     | 680                                                        | 780  | 900  | 1050 | 1350 | 1550 | 1800 | 1900 | 2000 | 2100 | ps                    |
| DQSCK Delta Long <sup>(6)</sup>      | t <sub>DQSCKDL</sub> | max     |                     | 920                                                        | 1050 | 1200 | 1400 | 1800 | 2100 | 2400 | -    | -    | -    | ps                    |
| DQS - DQ skew                        | t <sub>DQSQ</sub>    | max     |                     | 200                                                        | 220  | 240  | 280  | 340  | 370  | 400  | 500  | 600  | 700  | ps                    |
| Data hold skew factor                | t <sub>QHS</sub>     | max     |                     | 230                                                        | 260  | 280  | 340  | 400  | 450  | 480  | 600  | 750  | 1000 | ps                    |
| DQS Output High Pulse Width          | t <sub>QSH</sub>     | min     |                     | t <sub>CH</sub> (abs) - 0.05                               |      |      |      |      |      |      |      |      |      | t <sub>CK</sub> (avg) |
| DQS Output Low Pulse Width           | t <sub>QSL</sub>     | min     |                     | t <sub>CL</sub> (abs) - 0.05                               |      |      |      |      |      |      |      |      |      | t <sub>CK</sub> (avg) |
| Data Half Period                     | t <sub>QHP</sub>     | min     |                     | min(t <sub>QSH</sub> , t <sub>QSL</sub> )                  |      |      |      |      |      |      |      |      |      | t <sub>CK</sub> (avg) |
| DQ / DQS output hold time from DQS   | t <sub>QH</sub>      | min     |                     | t <sub>QHP</sub> - t <sub>QHS</sub>                        |      |      |      |      |      |      |      |      |      | ps                    |
| Read preamble <sup>(8)</sup>         | t <sub>RPRE</sub>    | min     |                     | 0.9                                                        |      |      |      |      |      |      |      |      |      | t <sub>CK</sub> (avg) |
| Read postamble <sup>(9)</sup>        | t <sub>RPST</sub>    | min     |                     | t <sub>CL</sub> (abs) - 0.05                               |      |      |      |      |      |      |      |      |      | t <sub>CK</sub> (avg) |
| DQS low-Z from clock <sup>(7)</sup>  | t <sub>LZ(DQS)</sub> | min     |                     | t <sub>DQSCK</sub> (MIN) - 300                             |      |      |      |      |      |      |      |      |      | ps                    |
| DQ low-Z from clock <sup>(7)</sup>   | t <sub>LZ(DQ)</sub>  | min     |                     | t <sub>DQSCK</sub> (MIN) - (1.4 * t <sub>QHS</sub> (MAX))  |      |      |      |      |      |      |      |      |      | ps                    |
| DQS high-Z from clock <sup>(7)</sup> | t <sub>HZ(DQS)</sub> | max     |                     | t <sub>DQSCK</sub> (MAX) - 100                             |      |      |      |      |      |      |      |      |      | ps                    |
| DQ high-Z from clock <sup>(7)</sup>  | t <sub>HZ(DQ)</sub>  | max     |                     | t <sub>DQSCK</sub> (MAX) + (1.4 * t <sub>DQSQ</sub> (MAX)) |      |      |      |      |      |      |      |      |      | ps                    |

LPDDR2 AC Timing Table (1,2)

| Parameter                                           | Symbol            | min max | min t <sub>CK</sub> | LPDDR2 |     |     |     |     |     |     |     |     |      | Unit                  |  |
|-----------------------------------------------------|-------------------|---------|---------------------|--------|-----|-----|-----|-----|-----|-----|-----|-----|------|-----------------------|--|
|                                                     |                   |         |                     | 1066   | 933 | 800 | 667 | 533 | 466 | 400 | 333 | 266 | 200  |                       |  |
| Write Parameters (3)                                |                   |         |                     |        |     |     |     |     |     |     |     |     |      |                       |  |
| DQ and DM input hold time (V <sub>ref</sub> based)  | t <sub>DH</sub>   | min     |                     | 210    | 235 | 270 | 350 | 430 | 450 | 480 | 600 | 750 | 1000 | ps                    |  |
| DQ and DM input setup time (V <sub>ref</sub> based) | t <sub>DS</sub>   | min     |                     | 210    | 235 | 270 | 350 | 430 | 450 | 480 | 600 | 750 | 1000 | ps                    |  |
| DQ and DM input pulse width                         | t <sub>DIPW</sub> | min     |                     | 0.35   |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
| Write command to 1st DQS latching transition        | t <sub>DQSS</sub> | min     |                     | 0.75   |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
|                                                     |                   | max     |                     | 1.25   |     |     |     |     |     |     |     |     |      |                       |  |
| DQS input high-level width                          | t <sub>DQSH</sub> | min     |                     | 0.4    |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
| DQS input low-level width                           | t <sub>DQSL</sub> | min     |                     | 0.4    |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
| DQS falling edge to CK setup time                   | t <sub>DSS</sub>  | min     |                     | 0.2    |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
| DQS falling edge hold time from CK                  | t <sub>DSH</sub>  | min     |                     | 0.2    |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
| Write postamble                                     | t <sub>WPST</sub> | min     |                     | 0.4    |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
| Write preamble                                      | t <sub>WPPE</sub> | min     |                     | 0.35   |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |

LPDDR2 AC Timing Table (1,2)

| Parameter                                                       | Symbol                  | min max | min t <sub>CK</sub> | LPDDR2 |     |     |     |     |     |     |     |     |      | Unit                  |  |
|-----------------------------------------------------------------|-------------------------|---------|---------------------|--------|-----|-----|-----|-----|-----|-----|-----|-----|------|-----------------------|--|
|                                                                 |                         |         |                     | 1066   | 933 | 800 | 667 | 533 | 466 | 400 | 333 | 266 | 200  |                       |  |
| CKE Input Parameters                                            |                         |         |                     |        |     |     |     |     |     |     |     |     |      |                       |  |
| CKE min. pulse width (high and low pulse width)                 | t <sub>CKE</sub>        | min     | 3                   | 3      |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
| CKE input setup time                                            | t <sub>ISCKE</sub> (10) | min     |                     | 0.25   |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
| CKE input hold time                                             | t <sub>IHCKE</sub> (11) | min     |                     | 0.25   |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
| Command Address Input Parameters (3)                            |                         |         |                     |        |     |     |     |     |     |     |     |     |      |                       |  |
| Address and control input setup time (V <sub>ref</sub> based)   | t <sub>IS</sub> (12)    | min     |                     | 220    | 250 | 290 | 370 | 460 | 520 | 600 | 740 | 900 | 1150 | ps                    |  |
| Address and control input hold time (V <sub>ref</sub> based)    | t <sub>IH</sub> (12)    | min     |                     | 220    | 250 | 290 | 370 | 460 | 520 | 600 | 740 | 900 | 1150 | ps                    |  |
| Address and control input pulse width                           | t <sub>IPW</sub>        | min     |                     | 0.40   |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
| Boot Parameters (10 MHz – 55 MHz) (13,14,15)                    |                         |         |                     |        |     |     |     |     |     |     |     |     |      |                       |  |
| Clock Cycle Time                                                | t <sub>CKb</sub>        | max     | -                   | 100    |     |     |     |     |     |     |     |     |      | ns                    |  |
|                                                                 |                         | min     | 81                  |        |     |     |     |     |     |     |     |     |      |                       |  |
| CKE Input Setup Time                                            | t <sub>ISCKEb</sub>     | min     | -                   | 2.5    |     |     |     |     |     |     |     |     |      | ns                    |  |
| CKE Input Hold Time                                             | t <sub>IHCKEb</sub>     | min     | -                   | 2.5    |     |     |     |     |     |     |     |     |      | ns                    |  |
| Address & Control Input Setup Time                              | t <sub>ISb</sub>        | min     | -                   | 1150   |     |     |     |     |     |     |     |     |      | ps                    |  |
| Address & Control Input Hold Time                               | t <sub>IHb</sub>        | min     | -                   | 1150   |     |     |     |     |     |     |     |     |      | ps                    |  |
| DQS Output Data Access Time<br>from CK/CK#                      | t <sub>DQSCkb</sub>     | min     | -                   | 2.0    |     |     |     |     |     |     |     |     |      | ns                    |  |
|                                                                 |                         | max     | 10.0                |        |     |     |     |     |     |     |     |     |      |                       |  |
| Data Strobe Edge to<br>Ouput Data Edge t <sub>DQSQb</sub> - 1.2 | t <sub>DQSQb</sub>      | max     | -                   | 1.2    |     |     |     |     |     |     |     |     |      | ns                    |  |
| Data Hold Skew Factor                                           | t <sub>QHSb</sub>       | max     | -                   | 1.2    |     |     |     |     |     |     |     |     |      | ns                    |  |
| Mode Register Parameters                                        |                         |         |                     |        |     |     |     |     |     |     |     |     |      |                       |  |
| MODE REGISTER Write command period                              | t <sub>MRW</sub>        | min     | 5                   | 5      |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |
| Mode Register Read command period                               | t <sub>MRR</sub>        | min     | 2                   | 2      |     |     |     |     |     |     |     |     |      | t <sub>CK</sub> (avg) |  |

LPDDR2 AC Timing Table <sup>(1,2)</sup>

| Parameter                                                                         | Symbol                      | min max    | min t <sub>CK</sub> | LPDDR2 SDRAM Core Parameters (16)                                                                                                |     |     |     |     |     |     |     |     |     | LPDDR2 |  |  |                       |  | Unit |
|-----------------------------------------------------------------------------------|-----------------------------|------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|--------|--|--|-----------------------|--|------|
|                                                                                   |                             |            |                     | 1066                                                                                                                             | 933 | 800 | 667 | 533 | 466 | 400 | 333 | 266 | 200 |        |  |  |                       |  |      |
| Read Latency                                                                      | RL                          | min        | 3                   | 8                                                                                                                                | 7   | 6   | 5   | 4   |     | 3   | 3   |     |     |        |  |  | t <sub>CK</sub> (avg) |  |      |
| Write Latency                                                                     | WL                          | min        | 1                   | 4                                                                                                                                | 4   | 3   | 2   | 2   |     | 1   | 1   |     |     |        |  |  | t <sub>CK</sub> (avg) |  |      |
| ACTIVE to ACTIVE command period                                                   | t <sub>RC</sub>             | min        |                     | t <sub>RAS</sub> + t <sub>RPab</sub> (with all-bank Precharge)<br>t <sub>RAS</sub> + t <sub>RPpb</sub> (with per-bank Precharge) |     |     |     |     |     |     |     |     |     |        |  |  | ns                    |  |      |
| CKE min. pulse width during Self-Refresh<br>(low pulse width during Self-Refresh) | t <sub>CKESR</sub>          | min        | 3                   |                                                                                                                                  |     |     | 15  |     |     |     | 15  |     |     |        |  |  | ns                    |  |      |
| Self refresh exit to next valid command delay                                     | t <sub>XSR</sub>            | min        | 2                   | t <sub>RRCaB</sub> + 10                                                                                                          |     |     |     |     |     |     |     |     |     |        |  |  | ns                    |  |      |
| Exit power down to next valid command delay                                       | t <sub>XP</sub>             | min        | 2                   |                                                                                                                                  |     |     | 7.5 |     |     |     | 7.5 |     |     |        |  |  | ns                    |  |      |
| LPDDR2-S4 CAS to CAS delay                                                        | t <sub>CCD</sub>            | min        | 2                   |                                                                                                                                  |     |     | 2   |     |     |     | 2   |     |     |        |  |  | t <sub>CK</sub> (avg) |  |      |
| Internal Read to Precharge command delay                                          | t <sub>RTP</sub>            | min        | 2                   |                                                                                                                                  |     |     | 7.5 |     |     |     | 7.5 |     |     |        |  |  | ns                    |  |      |
| RAS to CAS Delay <sup>(19)</sup>                                                  | t <sub>RCD</sub>            | Fast (min) | 3                   |                                                                                                                                  |     |     | 15  |     |     |     | 15  |     |     |        |  |  | ns                    |  |      |
|                                                                                   |                             | Typ (min)  | 3                   |                                                                                                                                  |     |     | 18  |     |     |     | 18  |     |     |        |  |  | ns                    |  |      |
| Row Precharge Time <sup>(19)</sup><br>(single bank)                               | t <sub>RPpb</sub>           | Fast (min) | 3                   |                                                                                                                                  |     |     | 15  |     |     |     | 15  |     |     |        |  |  | ns                    |  |      |
|                                                                                   |                             | Typ (min)  | 3                   |                                                                                                                                  |     |     | 18  |     |     |     | 18  |     |     |        |  |  | ns                    |  |      |
| Row Precharge Time <sup>(19)</sup><br>(all banks)                                 | t <sub>RPab</sub><br>4-bank | Fast (min) | 3                   |                                                                                                                                  |     |     | 15  |     |     |     | 15  |     |     |        |  |  | ns                    |  |      |
|                                                                                   |                             | Typ (min)  | 3                   |                                                                                                                                  |     |     | 18  |     |     |     | 18  |     |     |        |  |  | ns                    |  |      |
| Row Active Time                                                                   | t <sub>RAS</sub>            | min        | 3                   |                                                                                                                                  |     |     | 42  |     |     |     | 42  |     |     |        |  |  | ns                    |  |      |
|                                                                                   |                             | max        | -                   |                                                                                                                                  |     |     | 70  |     |     |     | 70  |     |     |        |  |  | us                    |  |      |
| Write Recovery Time                                                               | t <sub>WR</sub>             | min        | 3                   |                                                                                                                                  |     |     | 15  |     |     |     | 15  |     |     |        |  |  | ns                    |  |      |
| Internal Write to Read<br>Command Delay                                           | t <sub>WTR</sub>            | min        | 2                   |                                                                                                                                  |     |     | 7.5 |     |     |     | 10  |     |     |        |  |  | ns                    |  |      |
| Active bank A to Active bank B                                                    | t <sub>RBD</sub>            | min        | 2                   |                                                                                                                                  |     |     | 10  |     |     |     | 10  |     |     |        |  |  | ns                    |  |      |
| Four Bank Activate Window                                                         | t <sub>FAW</sub>            | min        | 8                   |                                                                                                                                  |     |     | 50  |     |     |     | 50  | 60  |     |        |  |  | ns                    |  |      |
| Minimum Deep Power Down Time                                                      | t <sub>DPD</sub>            | min        |                     |                                                                                                                                  |     |     | 500 |     |     |     | 500 |     |     |        |  |  | us                    |  |      |

Table 103 — LPDDR2 AC Timing Table

| Parameter                                    | Symbol                         | min<br>max | min<br>t <sub>CK</sub> | LPDDR2 |     |     |     |                          |     |      |     | Unit |     |
|----------------------------------------------|--------------------------------|------------|------------------------|--------|-----|-----|-----|--------------------------|-----|------|-----|------|-----|
|                                              |                                |            |                        | 1066   | 933 | 800 | 667 | 533                      | 466 | 400  | 333 |      | 266 |
| LPDDR2 Temperature De-Rating <sup>(17)</sup> |                                |            |                        |        |     |     |     |                          |     |      |     |      |     |
| Core Timings Temperature De-Rating           | t <sub>DQCK</sub><br>(Derated) | max        |                        | 5620   |     |     |     |                          |     | 6000 |     |      | ps  |
|                                              | t <sub>RCD</sub><br>(Derated)  | min        |                        |        |     |     |     | t <sub>RCD</sub> + 1.875 |     |      |     |      | ns  |
|                                              | t <sub>RC</sub><br>(Derated)   | min        |                        |        |     |     |     | t <sub>RC</sub> + 1.875  |     |      |     |      | ns  |
|                                              | t <sub>RAS</sub><br>(Derated)  | min        |                        |        |     |     |     | t <sub>RAS</sub> + 1.875 |     |      |     |      | ns  |
|                                              | t <sub>RP</sub><br>(Derated)   | min        |                        |        |     |     |     | t <sub>RP</sub> + 1.875  |     |      |     |      | ns  |
|                                              | t <sub>RRD</sub><br>(Derated)  | min        |                        |        |     |     |     | t <sub>RRD</sub> + 1.875 |     |      |     |      | ns  |

